License Manager

Complete Work Scope - One Integrated Business Application

Live local inspection + repository assessment | 31 October 2026

Purpose and boundary

This scope defines the work required for users to operate licence management, trade operations, financial control, accounting and bank reconciliation from one application. It is a proposed implementation plan, not evidence that features already work. Live inspection used an authenticated local session in read-only mode; no business records were created, edited, deleted, planned, recalculated or reconciled.

Live data signals requiring immediate attention

Reconciliation	1,043 BOEs; 317 import invoices; only 2 matched; 53 unmatched invoices; 1,699 duplicate-debit findings.	P0 data triage, classification, root-cause analysis and protected remediation workflow.
Data completeness	Dashboard reports 109 Missing DGFT records and 6 pending invoices.	Data-quality owner, required-field rules, remediation queue, source/import fixes.
Planning governance	E1 has 8 active rules and 2 validation issues; metadata says last update/modifier unavailable.	Show issue details, require audit metadata, approval and effective-version controls.
BOE operations	Live records expose inline invoice entry, merge, edit and delete actions; multi-licence BOEs occur.	Impact preview, permission/approval controls, idempotent writes and audit/reversal path.

Target operating model

A tenant-aware modular monolith remains appropriate: React provides user workflows; Django application services own all financial/licence calculations; PostgreSQL holds authoritative transactional data; Celery/Redis support durable jobs; reports use approved read models. A user should see a consistent company context, actionable work queues, clear calculation explanations, audit history and safe recovery on every screen.

Complete functional scope

Identity and administration	Sign in, profile, company/team switch, roles, approvals, activity review.	Tenant, membership, capability policy, object scope, MFA/session policy, admin audit.
Masters	Companies, counterparties, ports, HS/SION/product/rate masters with controlled changes.	Global-vs-tenant policy, validation, effective dating, imports, duplicate merge, change approval.
Licences	Create/import, review balances, documents, status, expiry, item details and lifecycle.	Canonical balance service, document integrity, state transitions, expiry work queue, audit/reversal.
Planning and restrictions	Configure rules, preview plan, allocate, explain result and correct exceptions.	Versioned rules, explicit groups, 2/3/5/10% pools, AU policy, reservations, locking, Decimal standard.
Allotments	Create, allocate/reverse, monitor shortfalls and capacity.	Idempotency, row locking, plan/restriction checks, allocation history and exception approval.
BOE	Capture/PDF import, lines/documents, licence linkage, transfer letters and corrections.	Structured extraction review, duplicate detection, unit/currency validation, immutable source/audit links.
Trade and invoices	Purchase/sale, lines, counterparties, payments, attachments and status.	Final-party validation, paired trade safety, invoice numbering, source-to-ledger bridge.
Reconciliation	Find/match BOE-invoice exceptions and show ownership/outcome.	Prioritised queues, deterministic candidates, differences, bulk triage, safe merge/reverse, audit.
Reports and exports	Operational, licence, planning and financial reporting.	Single service/read-model source, date/company filters, export parity and report versioning.

Accounting and bank reconciliation

Accounting must be implemented before claiming a Tally-level replacement. Existing commercial invoice/trade data becomes source data for controlled posting; it must not itself be treated as a general ledger.

Foundation	Legal entity/company boundary, chart of accounts, account groups, accounting periods, vouchers, opening balances.	Configured accounts and periods with an approved ownership and numbering policy.
Journal engine	Draft/approved/posted states, immutable journal lines, idempotent source references, reversals, period locks.	Every posted voucher satisfies debit = credit using documented Decimal/currency precision.
Subledgers	Sales, purchases, AR/AP, receipts, payments, credit/debit notes, advances and allocations.	Customer/supplier outstanding and ageing reconcile exactly to control accounts.
Tax/compliance	GST/TDS/export treatment, invoice data, tax summaries and audit exports.	Rules signed off by qualified Indian tax adviser and verified against approved examples.
Bank reconciliation	CSV/XLSX import, statement batches, normalized rows, dedupe, match suggestions, manual split/combine, adjustments and close/reopen.	Statement and book balances reconcile with explainable outstanding items and immutable history.
Migration	Tally mapping, opening balance/data import, exception reports and parallel reconciliation.	Migration reconciliation signed off; rollback/exception handling rehearsed.

Data-remediation workstream

Create a protected data-quality hub: issue type, severity, source system/file, affected entity, financial/licence impact, assignee, reason code, recommended action, approval, before/after values and audit event. Prioritise duplicate debits; BOE/invoice missing links; CIF/quantity mismatches; zero/missing quantities; unclassified products; missing DGFT fields; inconsistent company/product names; stale planning mappings; and orphaned documents. Use dry-run repair previews and reversible corrections. Do not mass-correct records before baseline exports and business approval.

UI/UX and workflow redesign

Dashboard	Useful counts, but only high-level signal; Missing DGFT and pending-invoice alerts lack visible remediation ownership.	Actionable queues, trend/age, data-quality score, owner/SLA and drill-down with safe filters.
Licence list	Long filter panel, opaque initial preselected statuses, wording defects and card density.	Saved views, filter chips, explainable defaults, compact table/cards toggle, clear reset, column preferences.
Planning	Validation issue count is visible but not actionable; metadata is unavailable.	Rule health drawer, impact simulation, version comparison, audit timeline, approval/review gates.
BOE	High-impact actions sit beside records; delete/merge/invoice action needs stronger guardrails.	Impact preview, reason/approval, undo/reversal, document panel and linked-record graph.
Reconciliation	Large exception volume; row actions require rapid but safe review.	Queue filters/priorities, candidate confidence, side-by-side evidence, bulk dry run, exception ownership.
Global system	Consistent navigation is now visible, but every module needs predictable loading/empty/error/success patterns.	Design tokens, accessible controls, command palette, responsive data grids, keyboard flow and Axe tests.

Important UX rules

Show currency, unit and date context beside every material number. Never hide a restrictive filter. Do not allow destructive or financially consequential actions without a clear impact statement, permission check, reason and audit record. Keep calculated balances read-only in the frontend; present backend explanation rather than duplicate calculations.

Delivery plan, tests and release gates

0 - Baseline	Confirm deployed version; isolate test database/Redis; golden data; data-quality/reconciliation register.	Environment, business SMEs.
1 - Containment	Tenant decision, permission/object scoping, critical data protection and regression tests.	Data classification and company model.
2 - SSOT	Balance, allocation and restriction service contracts; caller migration; report parity.	Golden examples, Decimal rules.
3 - UX foundation	Route/form/table/error primitives; dashboard/list/reconciliation work queues.	Stable APIs and workflow ownership.
4 - Restriction planning	Explicit pools, AU policy, versioned rules, plan/commit/concurrency/audit.	Regulatory/business decisions.
5 - Accounting	Ledger/subledgers/posting/reversal/periods, then financial reports.	Tenant and tax/CA review.
6 - Bank reconciliation	Import/adapters/matching/manual review/totals/audit/ledger linkage.	Posted accounting foundation and statement samples.
7 - Hardening	Performance, backups/restore, security, monitoring, accessibility, staging/release drills.	All prior contracts and test automation.

Minimum test matrix

Unit: balances, Decimal rounding, restriction resolution, journal balancing. Integration: locks, idempotency, reversals, imports, query scoping. API: validation and tenant/role negatives. E2E: licence -> plan -> allotment -> BOE -> trade -> reconciliation; invoice/payment -> journal -> bank match. Data: duplicate, missing, zero, invalid unit/currency, stale mapping. Nonfunctional: accessibility, responsive layouts, performance, restore/recovery and security scanning. A test is PASS only after execution in an isolated environment with recorded evidence.

Management decisions

Approve: internal multi-company versus external SaaS; company/tenant hierarchy; restriction calculation bases and AU meaning; accounting/tax scope; Tally migration requirement; supported bank formats; approval thresholds; historic-data correction policy; target scale and release authority. These decisions control the schedule and prevent building conflicting rules.

Assessment limitation

The live review confirms authenticated presentation and selected operational data, but did not perform mutations. Browser-driven functional tests, API write tests and financial posting tests must be run in a dedicated seeded test environment. The repository dependency index expected by AGENTS.md was unavailable. No application code, configuration, database record or production environment was changed.